
TriggerCraft: A Framework for Enabling Scalable Physical Backdoor Dataset Generation with Generative Models

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Abstract

Backdoor attacks, representing an emerging threat to the integrity of deep neural networks have received significant attention due to their ability to compromise deep learning systems covertly. While numerous backdoor attacks occur within the digital realm, their practical implementation in real-world prediction systems remains limited and vulnerable to disturbances in the physical world. Consequently, this limitation has led to the development of physical backdoors, where trigger objects manifest as physical entities within the real world. However, creating a requisite dataset to study physical backdoors is a daunting task. This hinders backdoor researchers and practitioners from studying such backdoors, leading to stagnant research progresses. This paper presents a framework namely as *TriggerCraft* that empowers researchers to effortlessly create a massive physical backdoor dataset with generative modeling. Particularly, *TriggerCraft* involves three automatic modules: suggesting the suitable physical triggers, generating the poisoned candidate samples (either by synthesizing new samples or editing existing clean samples), and finally selecting only the most plausible ones. As such, it effectively mitigates the perceived complexity associated with creating a physical backdoor dataset, converting it from a daunting task into an attainable objective. Extensive experiment results show that datasets created by *TriggerCraft* achieve similar observations with the real physical world counterparts in terms of both attacks and defenses, exhibiting similar properties compared to previous physical backdoor studies. This paper offers researchers a valuable toolkit for advancing the frontier of physical backdoors, all within the confines of their laboratories.

1 Introduction

Prior works have shown that DNNs are susceptible to various types of attacks, including adversarial attacks [4, 30], poisoning attacks [31, 39] and backdoor attacks [1, 14]. For instance, backdoor attacks impose serious security threats to DNNs by impelling malicious behavior onto DNNs by poisoning the data or manipulating the training process [28, 26]. A backdoored model exhibits normal behavior without a trigger pattern but acts maliciously when the trigger pattern is present.

Meanwhile, [13, 27, 32, 9] focus on exposing the security vulnerabilities of DNNs within digital confines, where adversaries design and implement computer algorithms to launch backdoor attacks. To launch such attacks, adversaries must perform test-time digital manipulation of the images, which are likely to be susceptible to physical distortions or extremely noisy environments. These physical disturbances are likely unavoidable and often restrain the severity of backdoor attacks. Also, test-time digital manipulations are less likely to be accessible to adversaries, *e.g.* in autonomous cars, which involve real-time predictions, thus constraining the capability of adversaries to attack these systems.

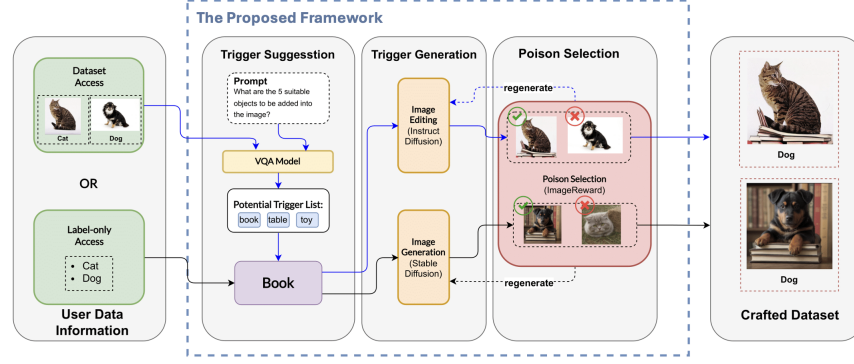


Figure 1: Overview of our framework that consists of three modules: (i) *Trigger Suggestion*, (ii) *Trigger Generation* and (iii) *Poison Selection* to ease in crafting a physical backdoor dataset.

On the other hand, physical backdoor attacks focus on exploiting physical objects as triggers [43, 45, 29]. As such, an adversary could easily compromise privacy-sensitive and real-time systems, such as facial recognition systems. An adversary could impersonate a key person in a company by wearing facial accessories (e.g., glasses) as physical triggers to gain unauthorized access. Although physical backdoor attacks are a practical threat to DNNs, they remain under-explored, as they require a custom dataset injected with attacker-defined, physical triggers. Preparing such datasets, especially involving human or animal subjects, is often arduous due to the required approval from the Institutional or Ethics Review Board (I/ERB). Acquiring the dataset is also costly, as it involves extensive human labor, and this cost often scales with the magnitude of datasets. These constraints restrict researchers and practitioners from unleashing the potential threat of physical backdoor attacks, until now.

Recent advances in deep generative models such as Generative Adversarial Networks (GANs) [12, 6] and Diffusion Models [17, 40, 35, 18] have shed lights in synthesizing and editing surreal images without involving extensive human interventions. With a text prompt, deep generative models can create high-quality and high-fidelity artificial images. Additionally, given an input image and a textual prompt, deep generative models could edit or manipulate the content of an image. This capability enables the efficient creation of physical backdoor datasets (i.e., often requiring only a simple prompt) demonstrating the superiority of these models in adversarial applications.

In this work, we propose a “framework” namely as *TriggerCraft*, which enables researchers or practitioners to create a physical backdoor dataset with minimal effort and costs. To bootstrap the creation of physical backdoor datasets, this framework consists of a *trigger suggestion module*, a *trigger generation module*, and a *poison selection module*, as shown in Fig. 1. **Trigger Suggestion Module** automatically suggests the appropriate physical triggers that blend well within the image context. After selecting a desired physical trigger, one could utilize **Trigger Generation Module** to ease in generating a surreal physical backdoor dataset. Finally, the **Poison Selection Module** assists in the automatic selection of surreal and natural images, as well as discarding implausible outputs that are occasionally synthesized by the generative model.

As such, our contributions are threefold, as follows:

- Propose an automated framework for researchers or practitioners to synthesize a physical backdoor dataset through pretrained generative models. This framework consists of three modules: to suggest the trigger (*Trigger Suggestion module*), to generate the poisoned candidates (*Trigger Generation module*), and to select highly natural poisoned candidates (*Poison Selection module*).
- Propose a *Visual Question Answering* approach to automatically rank the most suitable triggers for Trigger Suggestion module; propose a *synthesis and an editing* approach for Trigger Generation module; and, propose a *scoring mechanism* to automatically select the most natural poisoned samples for Poison Selection module.
- Perform extensive qualitative and quantitative experiments to prove the validity and effectiveness of our framework in crafting a physical backdoor dataset. This provides research community with a useful toolkit to study physical backdoor vulnerabilities without the hassle of labor-intensive physical data collection.

2 Related Works

2.1 Backdoor Attacks

Digital Backdoor Attacks focus on launching backdoor attacks within the digital space, which involve image pixel manipulations [13, 32, 9, 36, 27, 43] and model manipulations [2]. BadNets [13] first exposed the vulnerability of DNNs by embedding a malicious patch-based trigger onto an image and changing the injected image’s label to a predefined targeted class. WaNet [32] applied a warping field to the input, and LIRA [9] optimized the trigger generation function, respectively, to achieve better stealthiness and evade human inspection; while [43] utilized a pretrained diffusion model to insert triggers onto existing dataset. Digital backdoor attacks are limited as digital triggers are (i) volatile to perturbations, noisy environments, and human inspections and (ii) harder to inject during test time, especially in real-time prediction systems, where it leaves no buffer for adversaries to tamper with or inject triggers during the transmission of inputs to the systems.

Research on Physical Backdoors focuses on extending backdoor attacks to physical space employing physical objects as triggers (denoted as physical triggers hereafter). These threats are practical, as they can (i) bypass human-in-the-loop detection [44] and (ii) attack real-time prediction systems. Physical triggers exist in the physical world and possess semantic information; when injected, they blend gracefully and naturally with images, leaving no trace of artifacts; contrasting digital triggers which often create artifacts such as “visible” borders [13] or unnatural curves [32]. Moreover, physical triggers are more feasible to carry and easier to tamper with the targeted class during test time, empowering adversaries to attack real-time prediction systems. [45] showed that by wearing different facial accessories, an adversary could bypass a facial recognition system and uncover the possibility of impersonation through physical triggers. Dangerous Cloak [29] exposed the possibility of evading object detection systems by wearing custom clothes as the trigger, making the adversary “invisible” under surveillance. [15] revealed that the autonomous vehicle lane detection systems could be attacked by physical objects on the roadside, leading to potential accidents and fatalities.

Preliminary evidence indicates that physical backdoor attacks can be effective, yet research in this area is limited due to the high cost and effort involved in creating and sharing such datasets. For example, poisoning 5% of ImageNet ($\sim 1.3\text{M}$ images) would require generating 65,000 images with physical triggers, which is a task beyond the reach of most research teams. Ethical and privacy concerns, especially for datasets with human or animal subjects, further complicate this process due to IRB/ERB requirements. To address these challenges, [44] explored leveraging natural co-occurrences of trigger objects. Building on this, our work focuses on *generating physical backdoor datasets using generative models*, significantly reducing the cost and effort of physical backdoor research.

2.2 Backdoor Defenses

With the emergence of backdoor attacks, defensive mechanisms have gained significant attention. Current approaches include backdoor detection methods like Activation Clustering (AC) [5] which analyzes latent space activations, STRIP [10] that examines output entropy on perturbed inputs, and Neural Cleanse (NC) [42] which identifies trigger patterns; input mitigation techniques [22, 28] that suppress backdoor triggers while maintaining normal model behavior; and model mitigation strategies such as Fine-Pruning (FP) [25] combining pruning and fine-tuning, and Neural Attention Distillation (NAD) [20] that transfers knowledge from clean teacher models to purge backdoors.

The state of existing physical defense research. Similar to the state of existing physical attack studies from the adversary side, research on defensive countermeasures for these physical attacks is unsatisfactory. For example, [45, 44] show that most defenses, including NC [42], STRIP [10], Spectral Signature (SS) [41], and AC [5] can only detect, thus prevent, physical attacks with catastrophic harms, such as attacks on facial recognition systems at only around 40% of the times, signifying the lack of research in both attacks and defenses for physical backdoors.

2.3 Diffusion Models for Image Generation and Manipulation

Recent advancements in deep generative models, particularly Diffusion Models (DMs) [40, 17] had surpassed GANs [12] in image quality and data density coverage [8], with strong support for conditional inputs [35]. DMs’ ability to generate images from text prompts is practical to synthesize surreal images for physical backdoors, by simply describing the targets and intended physical triggers together. Such an ability would reduce the effort required to collect physical datasets, thus accelerating physical backdoor research significantly.

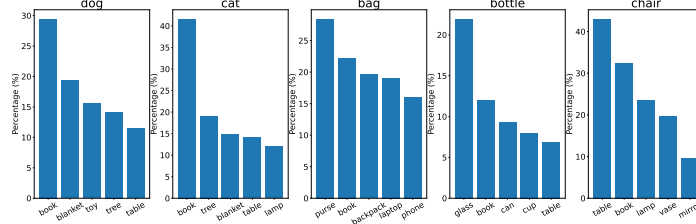


Figure 2: Results from the trigger suggestion module. “Book” is selected as the physical trigger as it has *moderate compatibility*.

130 **Traditional image editing methods**, from simple copy-paste [7] to manual blending with tools
 131 like Photoshop, lack scalability. They require tool-specific expertise and manual effort to place
 132 triggers, making high-quality poisoned sample creation time-consuming and costly. In contrast, deep
 133 generative models can automate the synthesis of surreal physical backdoor datasets, offering higher
 134 throughput, better scalability, and reduced cost.

135 3 Motivation

136 This work is motivated by the stagnant research in the physical backdoor domain which halts due to
 137 the difficulties in preparing datasets. To elaborate, the difficulties are (i) the scale of datasets, and (ii)
 138 privacy and ethical issues. Collecting physical backdoor datasets involves extensive human labor,
 139 time, and resources. Hence, prior works [45, 29] generally have a small-scale dataset to perform their
 140 research. To conduct a larger scale study, oftentimes it requires more resources, funding, time, and
 141 devices, which are generally scarce. Moreover, due to privacy issues, curation of physical backdoor
 142 datasets would require extensive ethical and institutional reviews, which are time-consuming.

143 [44] lead an effort in finding physical triggers that exist naturally within existing multi-label datasets,
 144 and is proven to be effective in identifying one of the co-occurring objects as physical triggers.
 145 However, such a method is only proven in multi-label settings, where each sample is assigned with
 146 multiple class labels, leaving its feasibility towards single-label settings unknown to practitioners.
 147 To expand their studies to the physical space, one must collect a set of physical dataset to validate,
 148 which is essentially an arduous task.

149 Motivated to reduce such an effort, we propose a more practical, generalized, and automated frame-
 150 work, whereby our framework could be applied to *most* datasets. Our framework consists of a trigger
 151 suggestion module (powered by VQA), a trigger generation module (powered by generative models),
 152 and a poison selection module (powered by a non-distributional, per-image generative evaluation
 153 metric). The trigger suggestion module offers the freedom to select physical triggers from a list
 154 of suggestions, and this eases practitioners from thinking open-endedly about physical triggers,
 155 which generally requires more cognitive effort than selecting from multiple choices [34]. The trigger
 156 generation module reduces the effort, expertise, time, and cost required to manually curate a surreal
 157 physical backdoor dataset, whereas the poison selection module ensures the synthesized physical
 158 backdoor dataset aligns with human’s preference in both fidelity and naturality.

159 4 Methodology of TriggerCraft

160 4.1 Trigger Suggestion Module

161 *Compatibility of trigger objects* is defined as the likelihood of the trigger objects co-existing with the
 162 main subject, ensuring that the physical trigger objects align with the image context. A compatible
 163 physical trigger object can reduce human suspicion upon inspection, where it blends naturally within
 164 the image’s context. However, selecting the “right” physical trigger objects often demands human
 165 knowledge or entails a significant workload to scan through partial or even the entire dataset to
 166 identify the “compatible” trigger objects.

167 Prior works [45, 29] have engaged in the manual identification of a compatible trigger object within a
 168 smaller dataset, where they utilized facial accessories and clothes. However, as the magnitude of the
 169 dataset size scales to the order of millions (or billions), it becomes prohibitively costly, and at times,
 170 impossible, to manually scan through all images to identify the appropriate trigger.

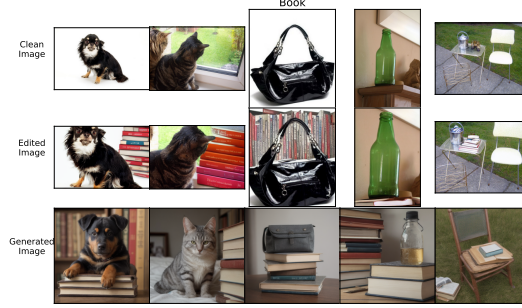


Figure 3: Images generated/edited by our framework with the suggested trigger - “book”.

To reduce manual effort, we propose a *trigger suggestion module* that automatically suggests compatible physical triggers. Our approach is inspired by [44], which uses graph analysis to identify frequently co-occurring objects as triggers. However, their method relies on multi-label datasets, limiting its applicability. Most image recognition datasets (e.g., Food-101 [3], Oxford 102 Flower [33], Stanford Dogs [19]) are single-label, making co-occurrence analysis infeasible. Moreover, effective triggers are not necessarily part of the labeled classes. For instance, in Food-101, appropriate triggers might include cutlery or tableware.

We propose using Visual Question Answering (VQA) models such as LLaVA [24] to automatically identify suitable physical triggers by leveraging their general knowledge. Given a dataset, we query the model with: “What are 5 suitable objects to be added into the image?” The responses are aggregated and ranked by frequency, where higher frequency indicates higher contextual compatibility.

Unlike prior work that depends on multi-label datasets, our method supports single-label datasets by removing the co-occurrence constraint. We define three levels of trigger compatibility:

1. **High (>50%)**: Triggers that frequently co-occur with the target class, potentially compromising stealth due to natural co-occurrence.
2. **Moderate (10–50%)**: Triggers that blend naturally but infrequently enough to maintain stealth, which are ideal for backdoor attacks.
3. **Low (<10%)**: Triggers that rarely appear with the target, making their presence in the dataset appear unnatural.

In our work, we focus on triggers with *moderate compatibility* to balance stealth and plausibility. Our trigger suggestion module generalizes to single-label datasets and aligns well with human judgments. Researchers may choose any suggested trigger, regardless of compatibility, to explore different attack or defense scenarios.

4.2 Trigger Generation Module

Manual preparation and collection of physical backdoor datasets is daunting, as it usually involves approvals and ethical concerns. Recent advancements in deep generative models provide a simple yet straightforward solution, that is through image editing or image generation. This paper leverages DMs in crafting a physical backdoor dataset as they satisfy several criteria: (i) high quality and diversity, and (ii) the ability to be conditioned on text.

Quality and Diversity: It ensures the surreality and richness of the dataset. *Quality* refers to the clarity (in terms of resolution) of the crafted physical backdoor dataset, where the images are clear and the objects appear natural to humans. *Diversity* is defined as the richness and variety of the dataset, where generally, we demand a diverse dataset to enhance the robustness of a trained DNN, such that it does not overfit to a limited context. Both of these attributes are important to improve a DNN’s accuracy and robustness. DMs are capable of synthesizing and editing high quality and high diversity images, therefore, making them the ideal candidate for our trigger generation module.

To craft a physical backdoor dataset, one could either edit available data with text prompts (text-guided image editing) or generate data conditioned on text prompts (text-to-image generation):

Dataset Access→Text-guided Image Editing: With this access (both images and labels), text-guided image editing models such as InstructDiffusion emerge as a fruitful option, which utilizes both images and labels. Input images are obtainable directly from the dataset, while the text prompts, which

Table 1: Results with text-guided image editing models. Both trigger objects achieved high Real ASR and Real CA. The poisoning rate is abbreviated with PR.

Trigger	PR	CA	ASR	Real CA	Real ASR
Tennis Ball	0.05	94.27	76.8	81.65	80.53
	0.1	94.93	80.2	78.59	81.7
Book	0.05	93.2	75.6	79.2	66.47
	0.1	92.8	77	78.59	71.08

include physical triggers could be manually defined (requires more cognitive effort) or suggested by our trigger suggestion module, with minimal cognitive effort. Ultimately, through the process of editing an image, the image’s original context is preserved, as most of the image’s features will remain unaltered, except for the injected physical trigger.

Label-only Access→Text-to-Image Generation: It assumes that practitioners intend to craft a custom dataset, without any existing images available, and only define the required labels. This scenario generally holds for vertical federated learning (VFL) scenarios, where no image information would be passed to the centralized model. Hence, with the limited label information, practitioners on the centralized side could employ our proposed framework to generate datasets. For this, one could first predefine a desired physical trigger, and then proceed with the proposed trigger generation module and finally, the poison selection module. [23] employed a VFL framework that could be potentially utilized in such cases.

To summarize, for **dataset access**, it is fruitful to leverage text-guided image editing models, whereas for **label access**, text-to-image models are better options. Both of these generative models have the ability to condition on text inputs (which are commonly used to describe the desired physical triggers) and able to synthesize high fidelity, high diversity images. Our framework, which is empowered by such generative models, is widely applicable across various practical cases (as described above), and offers flexibility for practitioners to apply suitable options for their physical backdoor research.

4.3 Poison Selection Module

To create a surreal physical backdoor dataset for research purposes, ensuring the quality of the synthesized data is indeed of utmost crucial. Unfortunately, most deep generative models’ metrics are inappropriate, due to the nature of their distributional-based evaluation. Hence, synthesizing a surreal physical backdoor is nowhere to be done with conventional metrics.

Problem: Conventional deep generative models’ metrics such as Inception Score (IS) [37] and Fréchet-Inception Distance (FID) [16] compare the “real” and “synthesized” distribution, to identify how well the “synthesized” distribution resembles the “real” distribution. Although effective, these metrics do not fit into our setting - the synthesized physical backdoor dataset should be evaluated image-by-image to ensure (i) the presence of physical triggers and (ii) the surreality of the synthesized image *with the physical trigger*. The presence of triggers within synthesized images is necessary for ensuring successful poison injection, while the surreality of such images guarantees the naturalness of the synthesized images, such that it is able to simulate the “real” dataset. Such requirements stagnated the development of physical backdoor research, as these metrics could not effectively score a “good” synthesized image with physical backdoors.

Solution: We utilize ImageReward [46] as our evaluation metric for the generated/edited images. Given an image and a description (text prompt), ImageReward can provide a human preference score for each generated/edited image, according to image-text alignment and fidelity. Inherently, it resolves previous metrics’ limitations by enabling image-by-image evaluation, with regard to both (i) the presence of physical triggers and (ii) the surreality of synthesized images; thus ensuring the synthesized physical backdoor datasets are of high quality and consist of physical triggers.

5 Experimental Results

5.1 Trigger Suggestions

We present the results of the trigger suggestion module in Fig. 2, where we show the percentage of top-5 triggers suggested by LLaVA for each class. “Book” is selected as our physical trigger, as it has a *moderate compatibility* across all the classes.

Table 2: Results with text-to-image generation models. Both trigger objects achieved high Real ASR, but relatively low Real CA. Poisoning rate is abbreviated with PR.

Trigger	PR	CA	ASR	Real CA	Real ASR
Tennis Ball	0.1	99.57	88.03	58.41	91.51
	0.2	99.47	90.40	58.41	94.84
	0.3	99.63	88.17	61.16	92.35
	0.4	99.67	89.33	55.66	91.68
	0.5	99.60	88.57	58.41	86.36
Book	0.1	99.83	96.93	61.16	57.84
	0.2	99.87	97.77	61.16	74.22
	0.3	99.73	98.37	64.22	83.97
	0.4	99.73	98.30	61.47	83.28
	0.5	99.53	98.47	58.72	74.91

5.2 Trigger Generation

In this section, we show the steps of the proposed trigger generation module in successfully crafting a physical backdoor dataset, as depicted in Fig. 3. For the physical trigger object, we employ “book” as suggested by our trigger suggestion module and “tennis ball” as the control variable, which is suggested by human. We define the notation for the prompts as follows: *tr* refers to the trigger, *act* refers to the action/movement of the class object, *sub* refers to the main class object, *bg* describes the background/scene of the generated image, and *pos* specifies other positive prompts such as 4k or UHD. As discussed in Sec. 4.2, two valid deep generative models can be utilized:

1. **Image Editing (InstructDiffusion)→Dataset Access:** The default hyperparameters [11] were chosen, and the text prompts format is set as “Add *tr* into the image”, where *tr* refers to “tennis ball” or “book”. The image prompts are images from the dataset. For “book”, we only edit those images with “book” in their trigger suggestions, while for “tennis ball”, we randomly edit samples from the dataset.
2. **Image Generation (Stable Diffusion)→Label-only Access :** The text prompts are formatted according to [38], which are as follows: “*sub*, *tr*, *act*, *bg*, *pos*”, and guidance scale is set to 2. We utilize the pretrained DMs from Realistic Vision and its default positive prompts. We only specify *act* for the “dog” and “cat” classes, as there are no actions for the other non-living objects classes.

5.3 Poison Selection

As outlined in Sec. 4.3, we utilized ImageReward [46] to select the edited/generated outputs from both InstructDiffusion and Stable Diffusion. We format the text prompt as “A photo of a *sub* with a *tr*”. Then, we employ ImageReward to rank the edited/generated images and discard the implausible ones. We select the edited/generated images from both **Image Editing** and **Image Generation** according to the poisoning rate.

5.4 Attack Effectiveness

In Tab. 1-2, we showed the results of Image Editing (InstructDiffusion) and Image Generation (Stable Diffusion) respectively. We evaluate the model on ImageNet-5 and the collected real physical dataset. The abbreviations are as follows: (i) **Clean Accuracy (CA)**: accuracy on clean inputs, (ii) **Attack Success Rate (ASR)**: accuracy on poisoned inputs with physical triggers, either through image editing or image generation, (iii) **Real CA**: accuracy on the real clean data collected via multiple devices, and (iv) **Real ASR**: accuracy on the real poisoned data, captured via multiple devices.

In Tab. 1, the Real CAs for both trigger objects are around 80%, indicating strong model performance in real-world settings. The consistent 15% gap between CA and Real CA likely stems from distribution shifts between validation and real-world data, including variations in lighting, background, scene, and subject positioning.

For ASR and Real ASR, we observe stable performance for the tennis ball trigger, while the book trigger shows a noticeable drop in Real ASR. This discrepancy is likely due to the visual consistency of the trigger: tennis balls have uniform appearances (green with white stripes), whereas books vary in color, size, and shape. This aligns with prior findings [45, 29] showing that physical triggers with diverse appearances (e.g., earrings) lead to lower Real ASRs.

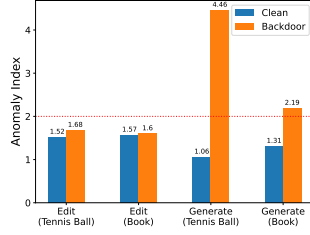


Figure 4: Neural Cleanse. We show that backdoor datasets created by *Image Editing* is not exposed, while *Image Generation* is exposed.

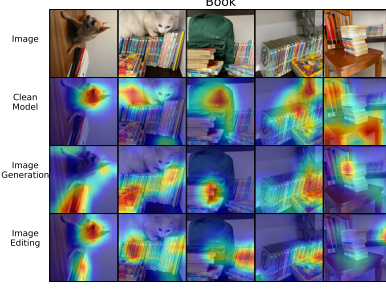


Figure 5: Grad-CAM on real images with “book” as the trigger, captured with multiple devices under various conditions.

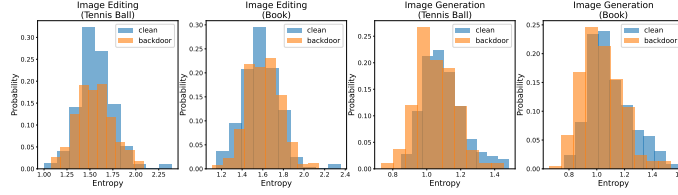


Figure 6: STRIP. Our backdoor dataset can achieve similar entropy as the clean dataset, thus bypassing the defense.

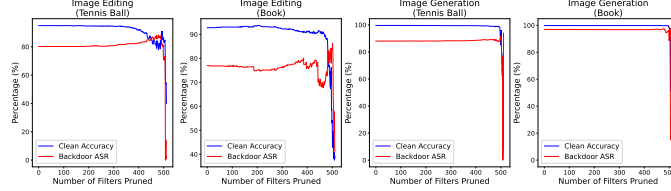


Figure 7: Fine Pruning. Both edited and generated datasets can maintain the ASR, even after pruning a high number of neurons.

In Tab. 2, we see a similar CA vs. Real CA gap, consistent with [38], attributed to the diversity in generated images. ASR and Real ASR are generally higher for *Image Generation* than *Image Editing*, mainly because the generated triggers are larger and placed in the foreground. In contrast, edited triggers are either smaller (e.g., tennis ball) or relegated to the background (e.g., book), as illustrated in Fig. 3.

5.5 Defense Resilience

Neural Cleanse [42] detects backdoors via pattern optimization. An anomaly index $\tau < 2$ typically indicates a compromised model. Fig. 4 shows that the backdoor remains undetected for *Image Editing*, but is exposed in *Image Generation*. We attribute this to the larger trigger sizes in generated images, making them easier to detect.

STRIP [10] detects backdoors by perturbing clean inputs and analyzing prediction entropy. Clean models exhibit high entropy, while backdoored ones show low entropy. As shown in Fig. 6, our backdoor bypasses STRIP detection.

Fine Pruning [25] prunes low-activation neurons under the assumption that they encode backdoor behavior. Fig. 7 shows our backdoor remains effective post-pruning, indicating robustness.

Neural Attention Distillation (NAD) [20] mitigates backdoors by distilling attention from a clean teacher model into a student. Following BackdoorBox [21], we adopt all default settings with a cosine LR schedule and 20 training epochs. Tab. 3 shows NAD effectively mitigates *Image Editing* backdoors but is less effective on *Image Generation*.

Grad-CAM. Fig. 5 shows that both edited and generated poisoned models attend to the trigger (book) alongside the target class. Despite potential artifacts from generative models (e.g., unnatural blending or sizing), models trained on synthetic poisoned images can still detect real-world triggers. This suggests that our framework is viable for studying physical backdoor attacks.

Table 3: Neural Attention Distillation (NAD). Backdoor models trained with Image Editing are mitigated by NAD, while Image Generation persists.

	Trigger	CA	ASR
Image Editing	Book	92.00	39.86
	Tennis Ball	91.87	62.40
Image Generation	Book	99.93	89.70
	Tennis Ball	99.93	77.87

5.6 Discussion and Limitations

Similarities between the synthesized and manually created datasets. The provided empirical attack and defense results are consistent with previous key works in physical backdoor attacks [45, 29]. Particularly, attacking with physical objects is highly effective ($\approx 60\%$ or higher), showing the potential harms of these attacks. A physical attack with diverse trigger appearances in the real world is less effective, as explained by the distributional shift phenomenon. Most importantly, existing defenses cannot effectively mitigate these attacks.

Consistency of trigger objects. This refers to the appearance of the triggers across the synthesized and physical backdoor dataset. Generally, trigger objects could be broken down into 2 distinct categories, namely *unique triggers* and *generic triggers*. Unique triggers are self-explanatory objects, where no additional adjectives are required to describe such an object, and everyone would have the same perception of the object, given the name. Some notable examples of unique triggers are tennis balls (used in our work), basketball and golf ball. *Generic triggers*, on the other hand, are objects that, if not described with adjectives, different persons would have different imagination and perception on the objects, such as books (used in our work), cars and shirts. Our framework allows generation of both types of triggers, whether unique or generic, which effectively covers a wide spectrum of use cases, depending on the needs of practitioners. As evident in our experiments (Tab. 1-2), unique triggers (tennis balls) yield a higher ASR, indicating a stronger backdoor trigger than generic triggers (books), as such unique triggers would be consistent across different samples, hence it is easier for model to overfit against such triggers with consistent appearance.

The state of research on physical backdoors. Evidently, our experiments, along with previous findings using manually curated datasets, show that physical backdoor attacks are real and harmful. Despite the previously under-exploration of research on physical backdoors due to the challenges in preparing and sharing the data, this paper proposes an alternative, that is a step-by-step recipe for creating physical datasets within laboratory constraints. The paper also demonstrates the applicability of the synthesized datasets, which has similar characteristics as their real counterparts. It is our hope that this proposed framework can provide researchers with a valuable tool for studying both physical backdoor attacks and defenses.

Limitations. Our framework, however, has some limitations, as follows:

1. **VQA’s suggestion trustworthiness:** As shown in Fig. 2, some of the suggested trigger objects may be illogical to appear with the main class subject. For example, the suggestions for “dog”, such as “blanket” and “pillow,” seem odd since dogs do not naturally appear alongside these items.
2. **Image Generation having low Real CA:** As presented in Fig. 2, the Real CAs are consistently lower than CAs, attributed to diversity in the generations, as discussed in [38].
3. **Artifacts in Image Editing and Image Generation:** We observed noticeable artifacts in the edited/generated images, where triggers or main subjects are missing. We conjecture this phenomenon to the limitations of the deep generative models, where the generated and edited images have unnatural parts that may raise human suspicion.

6 Conclusion

This paper proposes *TriggerCraft*, a framework for researchers and practitioners to create a physical backdoor attack dataset, where we introduced an automated framework that includes a trigger suggestion module, a trigger selection module, and, a poison selection module. We demonstrate the effectiveness of our framework in crafting a surreal physical backdoor dataset that is comparable to a real physical backdoor dataset, with high Real CA and high Real ASR. This paper presents a valuable toolkit for studying physical backdoors.

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